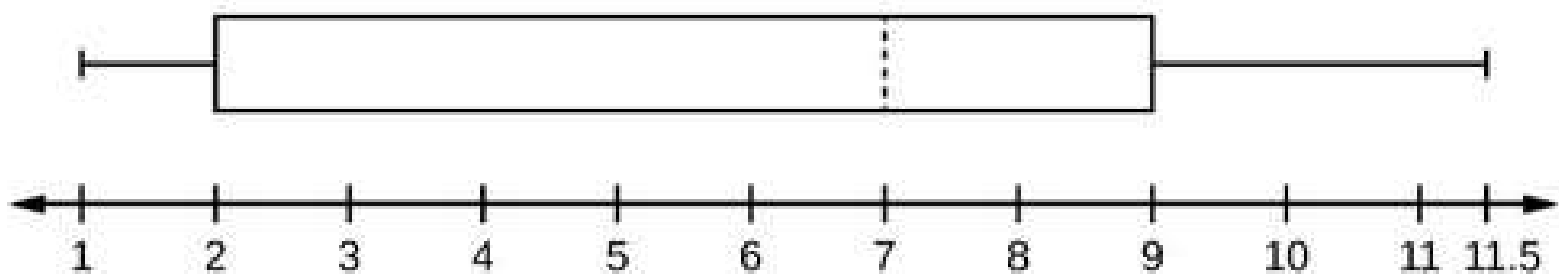


Box Plots

- Box plots (also called box-and-whisker plots or box-whisker plots) give a good graphical image of the concentration of the data. They also show how far the extreme values are from most of the data.
- A box plot is constructed from five values: the minimum value, the first quartile, the median, the third quartile, and the maximum value. We use these values to compare how close other data values are to them.

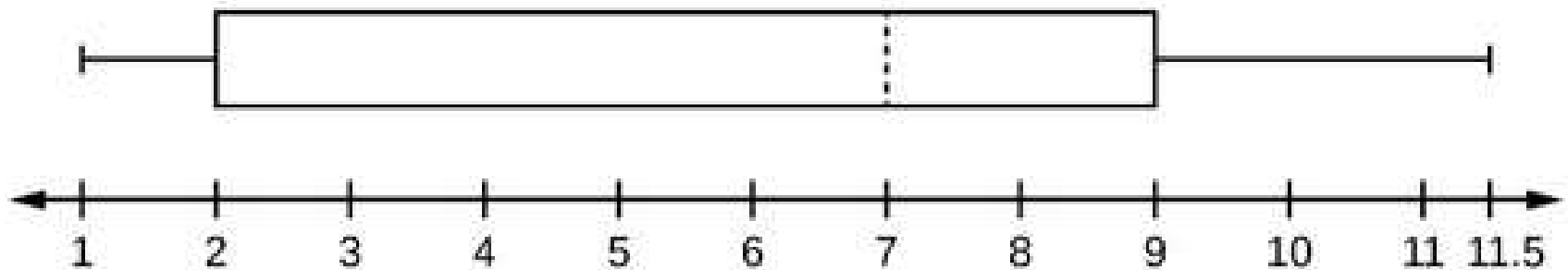


Box Plots

- Approximately the middle 50 percent of the data fall inside the box. The “whiskers” extend from the ends of the box to the smallest and largest data values. The median or second quartile can be between the first and third quartiles, or it can be one, or the other, or both. The box plot gives a good, quick picture of the data.

Example

- Consider, again, this dataset.
- 1 1 2 2 4 6 6.8 7.2 8 8.3 9 10 10 11.5
- The first quartile is two, the median is seven, and the third quartile is nine. The smallest value is one, and the largest value is 11.5. The following image shows the constructed box plot.



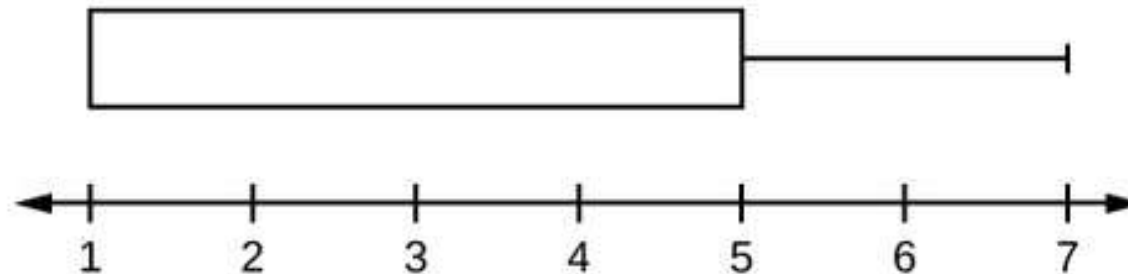
Example

- The following data are the heights of 40 students in a statistics class.
- 59 60 61 62 62 63 63 64 64 64 65 65 65 65 65 65 65 65 66 66 67 67 68 68 69 70 70 70 70 70 71 71 72 72 73 74 74 75 77
- Construct a box plot with the following properties.
- Minimum value = 59
- Maximum value = 77
- $Q1$: First quartile = 64.5
- $Q2$: Second quartile or median = 66
- $Q3$: Third quartile = 70

Box Plots

- For some sets of data, some of the largest value, smallest value, first quartile, median, and third quartile may be the same. For instance, you might have a data set in which the median and the third quartile are the same. In this case, the diagram would not have a dotted line inside the box displaying the median. The right side of the box would display both the third quartile and the median. For example, if the smallest value and the first quartile were both one, the median and the third quartile were both five, and the largest value was seven, the box plot would look like:

Box Plots



In this case, at least 25% of the values are equal to one. Twenty-five percent of the values are between one and five, inclusive. At least 25% of the values are equal to five. The top 25% of the values fall between five and seven, inclusive.

Measures of the Center of the Data

- **The Mean and Median**
- The “center” of a data set is also a way of describing location. The two most widely used measures of the “center” of the data are the mean (average) and the median. To calculate the mean weight of 50 people, add the 50 weights together and divide by 50. To find the median weight of the 50 people, order the data and find the number that splits the data into two equal parts. The median is generally a better measure of the center when there are extreme values or outliers because it is not affected by the precise numerical values of the outliers. The mean is the most common measure of the center.

Measures of the Center of the Data

- **Formulas to Find the Mean**
- To find the mean we add the values of a variable, and divide by the number of values. The letter used to represent the sample mean is an x with a bar over it (pronounced “x bar”) and the sample size is n:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

- The Greek letter μ (pronounced “mew”) represents the population mean, and the population size is N. One of the requirements for the sample mean to be a good estimate of the population mean is for the sample taken to be truly random.

Measures of the Center of the Data

Mean

The average of a set of values

$$\bar{x} = \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Example

- Find the mean, consider the sample:
- 1; 1; 1; 2; 2; 3; 4; 4; 4; 4; 4

$$\bar{x} = \frac{1 + 1 + 1 + 2 + 2 + 3 + 4 + 4 + 4 + 4 + 4}{11} = 2.7$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Measures of the Center of the Data

- **A Formula to Find the Median**
- You can quickly find the location of the median by using the expression below, which is the simplified form of the formula
- To find
- the 50th percentile, the Median.

$$i = \frac{k}{100}(n + 1) \xrightarrow{k=50} \frac{50}{100}(n + 1) = \frac{n + 1}{2}.$$

Measures of the Center of the Data

Median

- Divides the data into 2 equal parts, half below the median and half above
- If number of observations is odd - central value
- If even - halfway between the 2 central values

Mode

- Most frequently occurring data value

Example

- AIDS data indicating the number of months a patient with AIDS lives after taking a new antibody drug are as follows (smallest to largest):
- 3; 4; 8; 8; 10; 11; 12; 13; 14; 15; 15; 16; 16; 17; 17; 18; 21; 22; 22; 24; 24; 25; 26; 26; 27; 27; 29; 29; 31; 32; 33; 33; 34; 34; 35; 37; 40; 44; 44; 47;

Example

Calculate the mean and the median.

$$i = \frac{k}{100}(n + 1) = \frac{50}{100}(n + 1) = \frac{n + 1}{2}.$$

The calculation for the mean is:

$$\bar{x} = \frac{[3+4+(8)(2)+10+11+12+13+14+(15)(2)+(16)(2)+\dots+35+37+40+(44)(2)+47]}{40} = 23.6$$

To find the median, M , first use the formula for the location. The location is:

$$\frac{n+1}{2} = \frac{40+1}{2} = 20.5$$

Starting at the smallest value, the median is located between the 20th and 21st values (the two 24s):

3; 4; 8; 8; 10; 11; 12; 13; 14; 15; 15; 16; 16; 17; 17; 18; 21; 22; 22; 24; 24; 25; 26; 26; 27; 27; 29; 29; 31; 32; 33; 33; 34; 34;
35; 37; 40; 44; 44; 47;

$$M = \frac{24+24}{2} = 24$$

Measures of the Center of the Data

- **The Mode**
- Another measure of the center is the mode. The mode is the most frequent value. There can be more than one mode in a data set as long as those values have the same frequency and that frequency is the highest. A data set with two modes is called bimodal.

Example

- Statistics exam scores for 20 students are as follows:
- 50 53 59 59 63 63 72 72 72 72 72 76 78 81 83 84 84 84 90 93
- **Find the mode.**
- The most frequent score is 72, which occurs five times. Mode = 72.

Example

Calculating the Mean of Grouped Frequency Tables

$$\bar{x} = \frac{\sum_{i=1}^m x_i f_i}{\sum_{i=1}^m f_i}, \text{ where } \sum_{i=1}^m f_i = n$$

Example

A frequency table displaying professor Blount's last statistic test is shown. Find the best estimate of the class mean.

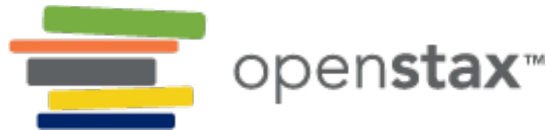
Grade Interval	Number of Students
50–56.5	1
56.5–62.5	0
62.5–68.5	4
68.5–74.5	4
74.5–80.5	2
80.5–86.5	3
86.5–92.5	4
92.5–98.5	1

Example

Midpoint	Number of Students
53.25	1
59.5	0
65.5	4
71.5	4
77.5	2
83.5	3
89.5	4
95.5	1

Or, calculate the total sum of products of values and frequencies, and divide by the sum of frequencies.

$$53.25(1)+59.5(0)+65.5(4)+71.5(4)+77.5(2)+83.5(3)+89.5(4)+95.5(1)=1460.25$$

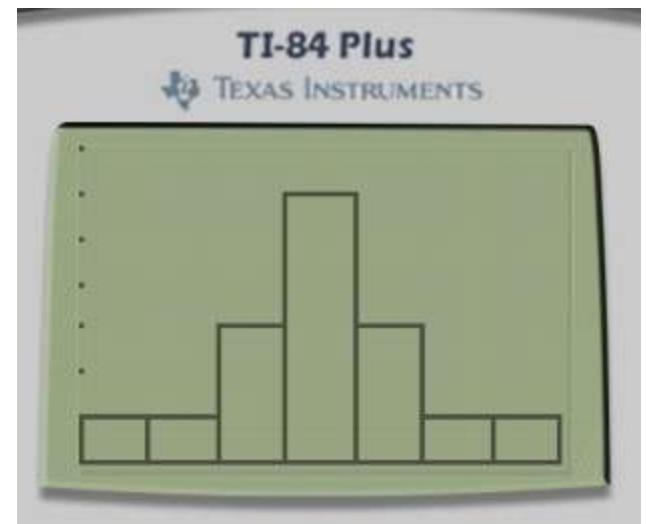


$$\bar{x} = \frac{\sum_{i=1}^m x_i f_i}{\sum_{i=1}^m f_i} = \frac{1460.25}{19}$$



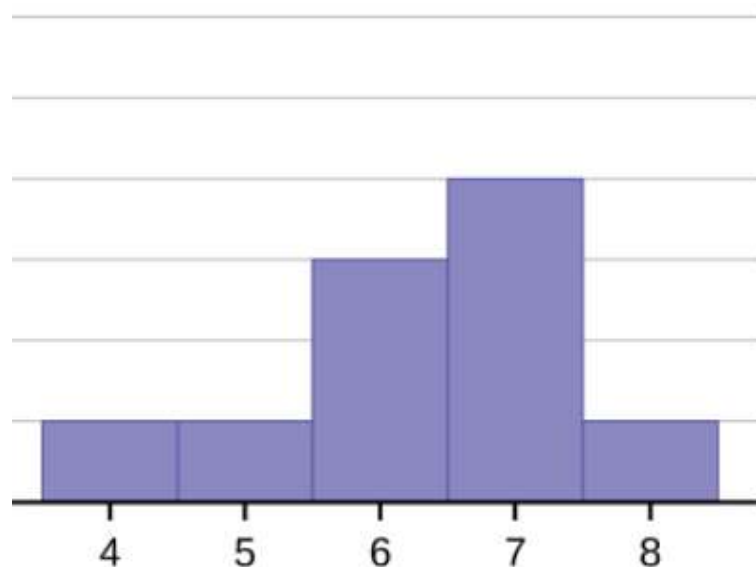
Skewness and the Mean, Median, and Mode

- Consider the following data set.
- 4; 5; 6; 6; 6; 7; 7; 7; 7; 7; 7; 8; 8; 8; 9; 10
The histogram displays a symmetrical distribution of data. A distribution is symmetrical if a vertical line can be drawn at some point in the histogram such that the shape to the left and the right of the vertical line are mirror images of each other. In a perfectly symmetrical distribution, the mean and the median are the same.



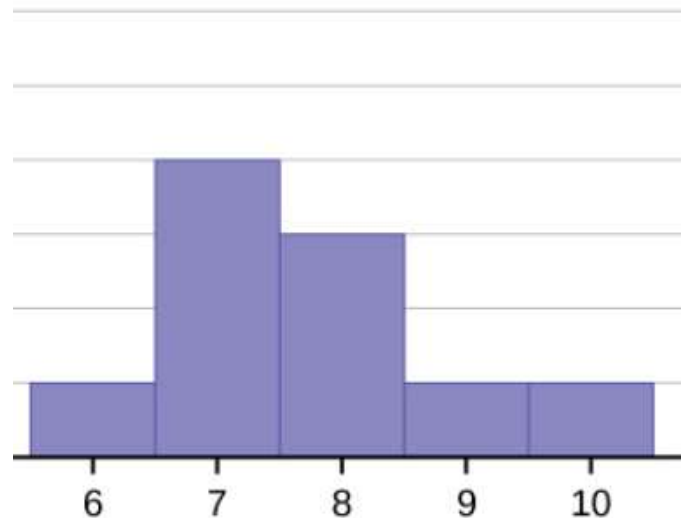
Skewness and the Mean, Median, and Mode

The histogram for the data: 4; 5; 6; 6; 6; 7; 7; 7; 7; 8 is not symmetrical. The right-hand side seems “chopped off” compared to the left side. A distribution of this type is called skewed to the left because it is pulled out to the left.



Skewness and the Mean, Median, and Mode

The histogram for the data: 6; 7; 7; 7; 7; 8; 8; 8; 9; 10, is also not symmetrical. It is skewed to the right.



Measures of the Spread of the Data

Standard deviation

- The average amount of variability in the data.
- Describes how far each score lies from the mean
- The larger the standard deviation, the more variability in the data

$$s = \sqrt{\frac{(x_i - \bar{x})^2}{n-1}}$$

Measures of the Spread of the Data

- **Example**
- Here is the list of ages of all close friends of Jasmine. The mean of the ages is 41.5.
- 41 38 39 45 47 41 44 41 37 42
- Every friend is either older than the average or younger. In other words, the ages deviate from the mean age. The question is: is there a number we can calculate that gives a measure for the overall deviation from the mean?

Measures of the Spread of the Data

The standard deviation and variance

41 38 39 45 47 41 44 41 37 42

The table below lists the friends with coded names. A good start is to calculate the deviations from the mean, and add them.

Ages	Deviation: $x - \mu$
A=41	$41 - 41.5 = -0.5$
B=38	$38 - 41.5 = -3.5$
C=39	$39 - 41.5 = -2.5$
D=45	$45 - 41.5 = 3.5$
E=47	$47 - 41.5 = 5.5$
F=41	$41 - 41.5 = -0.5$
G=44	$44 - 41.5 = 2.5$
H=41	$41 - 41.5 = -0.5$
L=37	$37 - 41.5 = -4.5$
M=42	$42 - 41.5 = 0.5$

The problem is that the sum will always be zero!

$$\sum_{i=1}^N (x_i - \mu) = 0$$

Measures of the Spread of the Data

The standard deviation and variance

41 38 39 45 47 41 44 41 37 42

The remedy is to square the deviations, or take absolute value of them to make them all positive. We'll square the deviations and add them.

Squares: $(x - \mu)^{2b}$

$$(-0.5)^2 = 0.25$$

$$(-3.5)^2 = 12.25$$

$$(-2.5)^2 = 6.25$$

$$(3.5)^2 = 12.25$$

$$(5.5)^2 = 30.25$$

$$(-0.5)^2 = 0.25$$

$$(2.5)^2 = 6.25$$

$$(-0.5)^2 = 0.25$$

$$(-4.5)^2 = 20.25$$

$$(0.5)^2 = 0.25$$

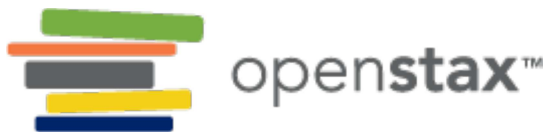
$$s^2 = \frac{\sum_{i=1}^n (x - \bar{x})^2}{n - 1} = \frac{88.5}{9} = 9.83$$

$$s = \sqrt{\frac{\sum_{i=1}^n (x - \bar{x})^2}{n - 1}}, \quad s = \sqrt{\frac{88.5}{9}} = 3.14$$



The standard deviation and variance

- A data value that is two standard deviations from the average is just on the borderline for what many statisticians would consider to be far from the average. Considering data to be far from the mean if it is more than two standard deviations away is more of an approximate “rule of thumb” than a rigid rule. In general, the shape of the distribution of the data affects how much of the data is further away than two standard deviations.
- Note that all ages of close friends of Jasmine are within 2 standard deviations
- Of the mean, between 35.56 and 47.44.
- 41 38 39 45 47 41 44 41 37 42



Example

- Use the following data (first exam scores) from Susan Dean's spring pre-calculus class:
- 33; 42; 49; 49; 53; 55; 55; 61; 63; 67; 68; 68; 69; 69; 72; 73; 74; 78; 80; 83; 88; 88; 88; 90; 92; 94; 94; 94; 94; 96; 100
- Create a chart containing the data, frequencies, relative frequencies, and cumulative relative frequencies to three decimal places.
- Calculate the following to one decimal place using a TI-83+ or TI-84 calculator:
 - The sample mean
 - The sample standard deviation
 - The median
 - The first quartile
 - The third quartile
 - IQR
- Construct a box plot and a histogram on the same set of axes. Make comments about the box plot, the histogram, and the chart.

Data	Frequency	Relative Frequency	Cumulative Relative Frequency
33	1	0.032	0.032
42	1	0.032	0.064
49	2	0.065	0.129
53	1	0.032	0.161
55	2	0.065	0.226
61	1	0.032	0.258
63	1	0.032	0.29
67	1	0.032	0.322
68	2	0.065	0.387
69	2	0.065	0.452
72	1	0.032	0.484
73	1	0.032	0.516
74	1	0.032	0.548
78	1	0.032	0.580
80	1	0.032	0.612
83	1	0.032	0.644
88	3	0.097	0.741
90	1	0.032	0.773
92	1	0.032	0.805
94	4	0.129	0.934
96	1	0.032	0.966
100	1	0.032	0.998

Note that relative frequencies don't add to 1 because of rounding errors.



Example

- Use the following data (first exam scores) from Susan Dean's spring pre-calculus class:
- 33; 42; 49; 49; 53; 55; 55; 61; 63; 67; 68; 68; 69; 69; 72; 73; 74; 78; 80; 83; 88; 88; 88; 90; 92; 94; 94; 94; 94; 96; 100
- Calculate the following to one decimal place using a TI-83+ or TI-84 calculator:
 - The sample mean
 - The sample standard deviation
 - The median
 - The first quartile
 - The third quartile
 - IQR

Comparing Values from Different Data Sets

- The standard deviation is useful when comparing data values that come from different data sets. If the data sets have different means and standard deviations, then comparing the data values directly can be misleading. In such cases we calculate the z score for each value.
- For each data value, calculate how many standard deviations away from its mean the value is. Find the distance between the value and the mean, then divide by standard deviation to find how many standard deviations fit in the distance. Use the formulas below for values from a population or a sample.

Comparing Values from Different Data Sets

For values from a sample: $z = \frac{x - \bar{x}}{s}$

For values from a population: $z = \frac{x - \mu}{\sigma}$

Comparing Values from Different Data Sets

Two students, John and Ali, from different high schools, wanted to find out who had the highest GPA when compared to his school. Which student had the highest GPA when compared to his school?

Student	GPA	School Mean GPA	School Standard Deviation
John	2.85	3.0	0.7
Ali	77	80	10

For values from a population: $z = \frac{x - \mu}{\sigma}$

$$z_{John} = \frac{2.85 - 3.0}{0.7} = -0.21$$

$$z_{Ali} = \frac{77 - 80}{10} = -0.3$$

John has the better GPA when compared to his school because his GPA is 0.21 standard deviations below his school's mean while Ali's GPA is 0.3 standard deviations below his school's mean.